

Michael Schramm

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Research Area

I work at the intersection of environmental science and policy by facilitating water quality planning efforts with state agencies and local stakeholders. I provide expertise in water quality modeling and assessment through the use of GIS and open source programming tools. My primary interest revolves around water policy and the evaluation of implementation effectiveness. I'm especially interested in leveraging open data and open source analytic tools to develop toolkits and resources for evaluating the environmental outcomes of policy implementation.

Skills

data analysis, GIS, grant writing, policy analysis, Python, R, reproducible research, stakeholder facilitation, water quality modeling and assessment, watershed planning

Research Experience

TEXAS A&M AGRI LIFE RESEARCH
Texas Water Resources Institute

MAY 2016 - CURRENT

- Research Specialist IV (Sept. 2023 - current)
- Research Specialist III (Aug. 2019 - Aug. 2023)
- Research Associate (May 2016 - Aug. 2019)

Leads watershed management and water quality assessment projects. Responsible for cultivating and maintaining relationships with project sponsors, developing and securing external funding, and successfully executing projects. Develops internal and external facing open-source tools for watershed planning, data analysis, and data sharing.

OAK RIDGE NATIONAL LABORATORY/
OAK RIDGE ASSOCIATED UNIVERSITIES
Environmental Sciences Division

FEB. 2014 - FEB. 2016

- Research Associate (Feb. 2014 - Feb. 2016)

Collaborator on projects that improved the understanding of environmental impacts and mitigation of hydropower technologies. Work included database development, statistical analysis, experiments, and publication.

UNIVERSITY OF DELAWARE
Center for Energy and Environmental Policy

SEPT. 2012 - JUNE 2013

- Graduate Research Assistant (Sept. 2012 - June 2013)

Provided research, data analysis, and conducted interviews for environmental policy analysis reports commissioned by the Delaware General Assembly. Policy analysis work focused on green infrastructure promotion and revitalization of brownfield sites.

Education

2011 - 2013	University of Delaware
Master of Energy and Environmental Policy	Newark, DE
2010 - 2011	University of North Carolina - Wilmington
B.A. Environmental Studies	Wilmington, NC
1999 - 2004	University of North Carolina - Wilmington
B.S. Biology	Wilmington, NC

Relevant Peer Reviewed Articles

Fourteen published articles, h-index = 8, i₁₀-index = 7

- Schramm, M. P., Kikoyo, D., Wright, J., & Jain, S. (2024). A meta-analysis of the impacts of best management practices on nonpoint source pollutant concentration. *Frontiers in Water*, 6(1397615). <https://doi.org/10.3389/frwa.2024.1397615>
- Schramm, M. (2023). Assessing linkages between watershed nutrient loading and water quality in a subtropical estuary with generalized additive models. *PeerJ*. <https://doi.org/10.7717/peerj.16073>
- Schramm, M., Gitter, A., & Gregory, L. (2022). Total Maximum Daily Loads and *Escherichia coli* trends in Texas freshwater streams. *Journal of Contemporary Water Research & Education*, 176, 36–49. <https://doi.org/10.1111/j.1936-704X.2022.3374.x>
- Schramm, M. P. (2021). Estimating statistical power for detecting long term trends in surface water *Escherichia coli* concentrations. *Texas Water Journal*, 12(1), 140–150. <https://doi.org/10.21423/twj.v12i1.7126>

Relevant Technical Reports

- Schramm, M. (2023). *Texas Coastal Nutrient Input Repository - Task 3 Report Lavaca Bay Water Quality Responses to Nutrient Loading* (Technical Report TR-546). Texas Water Resources Institute. https://twri.tamu.edu/media/6194/tr_546.pdf
- Schramm, M. (2023). *Texas Coastal Nutrient Input Repository - Task 4 Report Statistical Models for Nutrient Loading into Lavaca Bay* (Technical Report TR-543). Texas Water Resources Institute. https://twri.tamu.edu/media/6170/tr_543.pdf
- Schramm, M. P., deVilleneuve, S., Jain, S., Berthold, A., & Mohandass, U. (2019). *Carancabua Bay Watershed Protection Plan* (Technical Report TR-514). Texas Water Resources Institute. https://twri.tamu.edu/media/6170/tr_514.pdf

[//twri.tamu.edu/publications/technical-reports/2019-technical-reports/tr-514/](https://twri.tamu.edu/publications/technical-reports/2019-technical-reports/tr-514/)
Schramm, M. P., Berthold, A., & Entwistle, C. (2017). *Tres Palacios Creek Watershed Protection Plan* (Technical Report TR-500). Texas Water Resources Institute. <https://twri.tamu.edu/media/1449/tr-500.pdf>

Selected Grants and Contracts

Total external funding as PI or co-PI: \$3.5M

- 2025 - 2027 Texas Commission On Environmental Quality TMDL Program: Tres Palacios Bay Recreational Beach Watershed Characterization. Role: Co-PI; PI: Lucas Gregory. Amount: \$272,688.
- 2023 - 2024 Texas Commission On Environmental Quality TMDL Program: Research Methods to Address the Tres Palacios Bay Recreational Beach Bacteria Impairment. Role: Co-PI, PI: Lucas Gregory. Amount: \$62,603.
- 2020 - 2021 Texas Commission On Environmental Quality TMDL Program: Basins Approach to Address Bacterial Impairments in Basins 15, 16, 17 (FY19); Role: Project Manager, PI: Allen Berthold. Amount: \$36,212.
- 2018 - 2021 Section 319 Nonpoint Source Grant: Tres Palacios On-Site Sewage Facilities Remediation; Role: Grant Writer, PI: Allen Berthold. Amount: \$327,361.
- 2018 - 2021 Section 319 Nonpoint Source Grant: Tres Palacios WPP implementation; Role: Grant Writer, PI: Allen Berthold. Amount: \$355,800.
- 2019 - 2020 Texas Commission On Environmental Quality TMDL Program: Basins Approach to Address Bacterial Impairments in the Lower Neches Basin (FY20); Role: Project Manager, PI: Lucas Gregory. Amount: \$135,790.
- 2019 - 2020 Texas Commission On Environmental Quality TMDL Program: Basins Approach to Address Bacterial Impairments in Basins 15, 16, 17 (FY20); Role: Project Manager, PI: Allen Berthold. Amount: \$51,888.
- 2017 - 2019 Texas General Land Office Coastal Management Program: Coordinating Implementation of the Tres Palacios Watershed Protection Plan; Role: Grant Writer, PI: Allen Berthold. Amount: \$95,816.
- 2017 - 2018 Texas Commission On Environmental Quality TMDL Program: Basins Approach to Address Bacterial Impairments in Basins 15, 16, 17 (FY18). Amount: \$220,166.
- 2016 - 2017 Texas Commission On Environmental Quality TMDL Program: Basins Approach to Address Bacterial Impairments in Basins 15, 16, 17 (FY17); Role: Project Manager, PI: Allen Berthold. Amount: \$292,699.

Awards

Universities Council on Water Research (UCOWR), Journal Of Contemporary Water Research and Education Paper of the Year (2023)

Center for Energy and Environmental Policy Leadership Award (2013)

North Carolina Department of Transportation Environmental Service Scholarship (2011)

Service

REVIEW COMMITTEES

2017 - current: TWRI Mills Scholarship Program

2017 - current: USGS Graduate Research Program

2018 - 2022: Texas Hill Country Headwaters Conservation Initiative Regional Conservation Partnership Program

2021 - 2021: Texas A&M AgriLife, Texas A&M AgriLife Extension, and Texas A&M Engineering Station Water Exceptional Item Programs of Excellence

2019 - 2020: Texas A&M AgriLife, Texas A&M AgriLife Extension, and Texas A&M Engineering Station Water Seed Grant Initiative

PEER REVIEW

Aquatic Sciences, Journal of Open Source Software, R Journal